Penn Engineering
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FA26

Special Seminar on
Multi-Modal AI

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Lecture 14:
Connecting Modalities to LLMs
Special Seminar on Multi-Modal AI

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Representations in Multimodal Models

- In language models: every task is next-token-prediction (NTP)
- How to achieve this for different modalities?
- Tokenization: represent each modality as a sequence
- Today: make these representations usable by a pretrained LLM

Goal of This Lecture

Integrate pretrained perception and language models
to answer questions about sensory inputs

Input: image / audio + a task instruction
Output: a text response

Can We “Pretrain” Everything Together?

In short: yes.

In practice, a lot of limitations:

- Visual representation learning usually doesn’t work well with NTP; therefore, it may take significantly longer than separated training
- Engineering perspective: reuse computation as much as possible, avoid the problem of data mixture

Today’s Agenda

Part 1: Principles of Multimodal Architectures

Part 2: Principles of Multimodal Alignment

Part 3: Full Pipelines for VLM Instruction Following

The Simplest Interface: Continuous Prefixes



$$Z = C_\phi\left(E_\psi(X)\right), \qquad p(y \mid X, q) = \prod_{t=1}^{T} p_\theta\left(y_t \mid y_{<t}, Z, q\right)$$

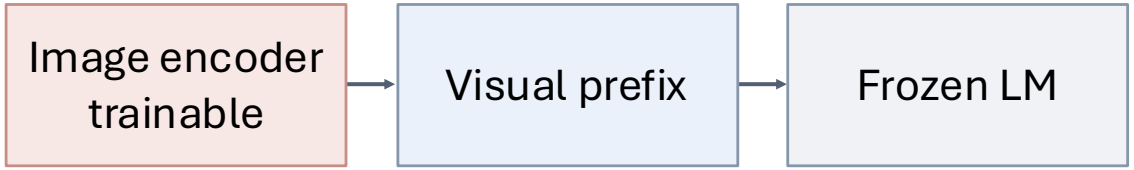
Advantage: Gradient flow

Frozen weights can still pass gradients to the connector

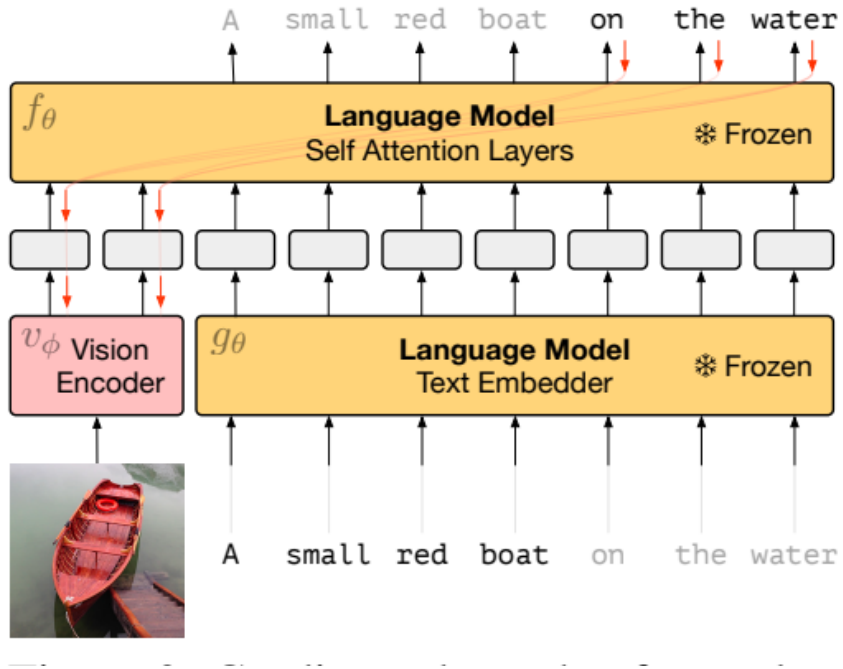
Simplest Idea: BP Through a Frozen Language Model

Key idea

Treat visual features as a “continuous prompt” to the language model



Use caption likelihood to train the visual encoder



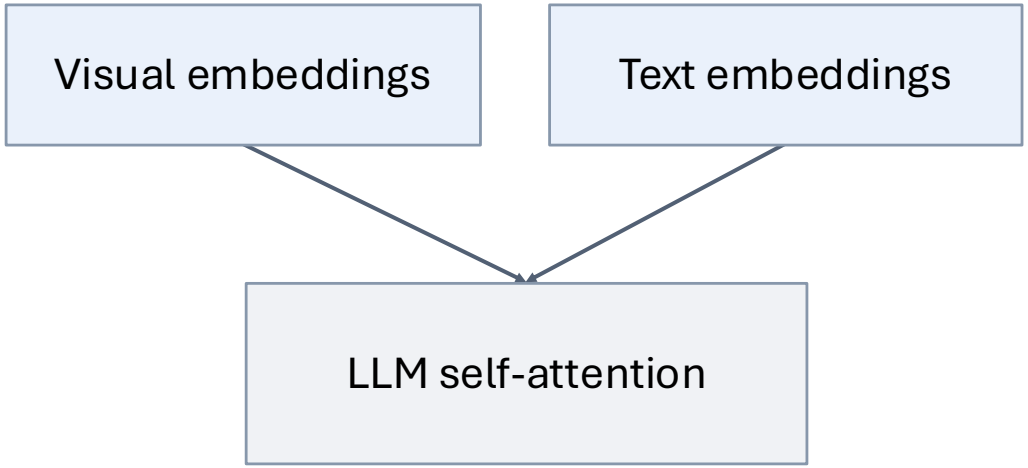
Original Figure 2: gradients through the frozen LM

Limitation: Not very good performance

Tsimpoukelli et al., 2021 · https://arxiv.org/abs/2106.13884

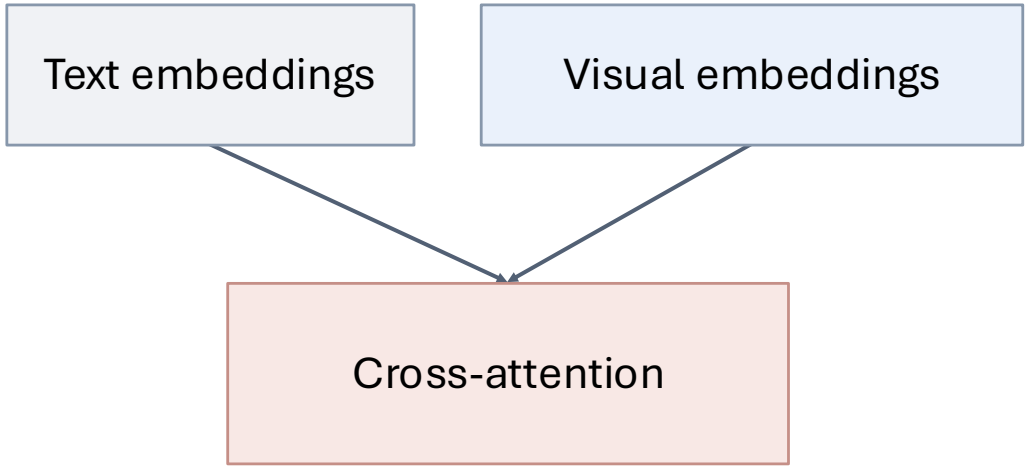
Architectures

Input-level fusion



Visual tokens as a prefix
No new attention type is required

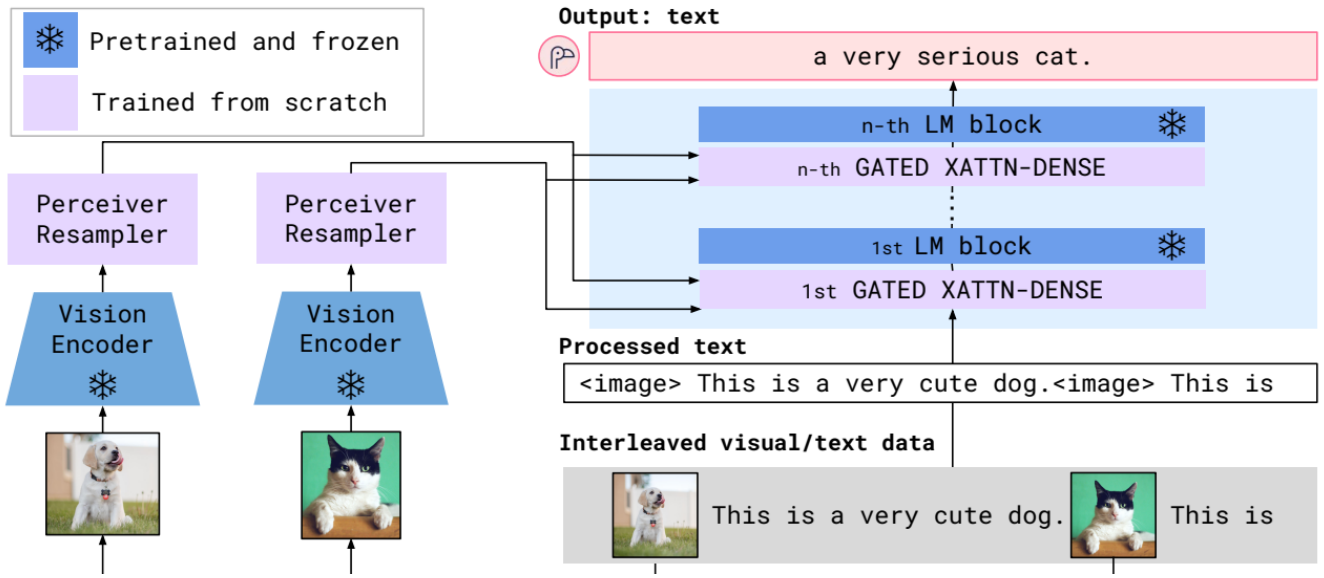
Cross-attention fusion



Visual features as a separate “KV store”
Need new cross-attention layers

Architecture synthesis: Frozen, Flamingo, BLIP-2, LLaVA

Flamingo: A Cross-Attention VLM Architecture



Original Figure 3: resampler plus gated cross-attention

Key idea

Reuse pretrained perception and language; learn the cross-attention

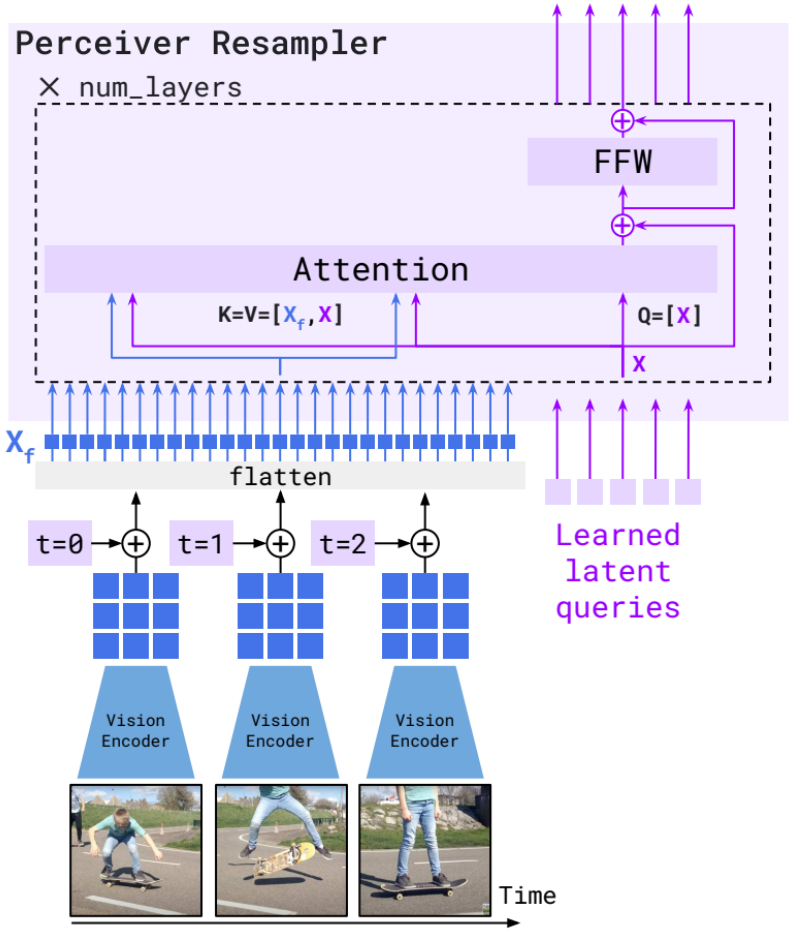
What is trained?

Resampler + inserted attention/dense blocks

Vision encoder and LM stay frozen

Alayrac et al., 2022 · https://arxiv.org/abs/2204.14198

Flamingo: Perceiver Resampler



Original Figure 5 (5 slots drawn; model uses 64)

1. Visual Embeddings

Encode each frame, add learned time embeddings, then flatten all spatial locations across frames

2. Learned latent queries

Initialize 64 learned vectors
Queries come from the current latents
Keys / values come from visual features + latents
Same strategy as TiTok!

3. Output

Repeat residual attention + feed-forward 6 times
Return 64 input-dependent tokens to the LLM

Variable-size visual input, fixed-size memory for language cross-attention

Flamingo: Gating and Multimodal Training

$$\tilde{h} = h + \tanh(\alpha)\,\mathrm{CrossAttn}(h, Z, Z), \qquad \alpha_0 = 0$$

$$\mathcal{L} = \sum_d \lambda_d \; \mathbb{E}_{(X,y)\sim D_d}\left[-\sum_t \log p(y_t \mid y_{<t}, X_{\le t})\right]$$

Stable insertion

Zero gates initially preserve the frozen LM's computation

Image-causal masking

Each text token directly attends to the most recent preceding image

Alayrac et al., 2022 · https://arxiv.org/abs/2204.14198

Principles of Multimodal Architectures

- A language model can be conditioned on continuous vectors directly
- Prefix-token models use self-attention; external-embedding models uses new cross-attention layers
- Still, need “compression” before attention; It’s basically tokenization!

Today’s Agenda

Part 1: Principles of Multimodal Architectures

Part 2: Principles of Multimodal Alignment

Part 3: Full Pipelines for VLM Instruction Following

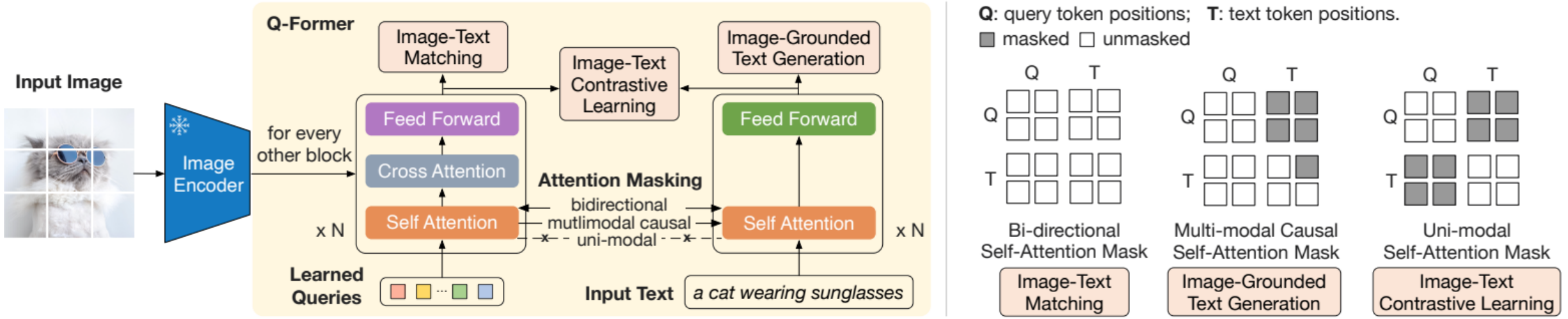
Learning The Connector/Compressor

- Suppose we freeze both the image encoder and the language model
- The queries must select useful visual evidence and make it decodable by the LLM
  - For example, remove some texture details
  - And align the representation space of “vision” and “language”

The BLIP Solution:

- BLIP-2 separates these into two learning problems
- First learn text-relevant visual features; then connect those features to the generator

BLIP-2: Learning Text-Relevant Visual Queries



Original Figure 2: Q-Former and objective-specific attention masks

Li et al., 2023 · https://arxiv.org/abs/2301.12597

BLIP-2: What Are the Learned Queries?

32 learned slots collect visual information

$$Q^{(0)} \in \mathbb{R}^{32\times 768}, \qquad Z(I) = \mathrm{QFormer}\left(Q^{(0)}, V(I)\right)$$

Every image starts with the same learned vectors
Their output states change with the image

Slot identity and spatial position are different

A query can attend to many image locations
Its index does not specify an image patch or an object category

Text uses BERT position embeddings
Queries have learned identities, without those text position embeddings

Li et al., BLIP-2 (2023) · arxiv.org/abs/2301.12597

BLIP-2: Information Flow in Q-Former

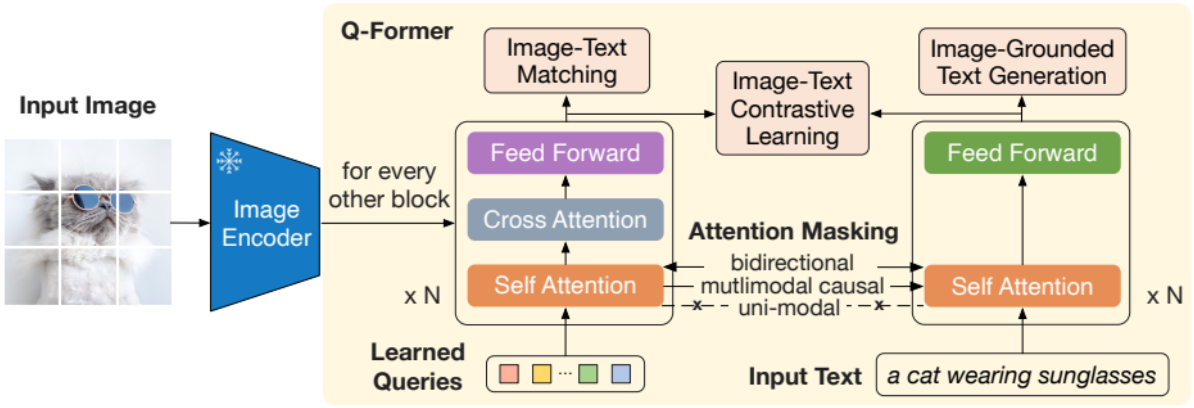


Figure 2 detail: query and text pathways

Shared self-attention

Queries and text share attention weights
The task mask controls the attention

Visual cross-attention

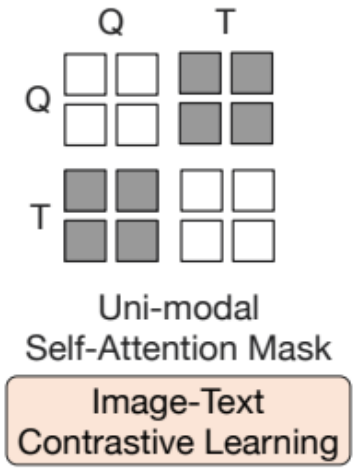
Only query states read image features
Cross-attention appears every other block

$$\mathrm{Attn}(H_Q, V) = \mathrm{softmax}\left(\frac{(H_Q W_Q)(V W_K)^\top}{\sqrt{d_k}}\right) V W_V$$

Li et al., BLIP-2 (2023) · arxiv.org/abs/2301.12597

BLIP-2: Contrastive Learning vs Matching

ITC: retrieve the paired caption



Encode image and text without cross-modal exchange

Compare normalized features against in-batch negatives

ITM: check a candidate pair



Allow queries and text to interact bidirectionally

Classify matched vs unmatched using hard negative pairs

Li et al., BLIP-2 (2023) · arxiv.org/abs/2301.12597

BLIP-2: Generation Through the Queries

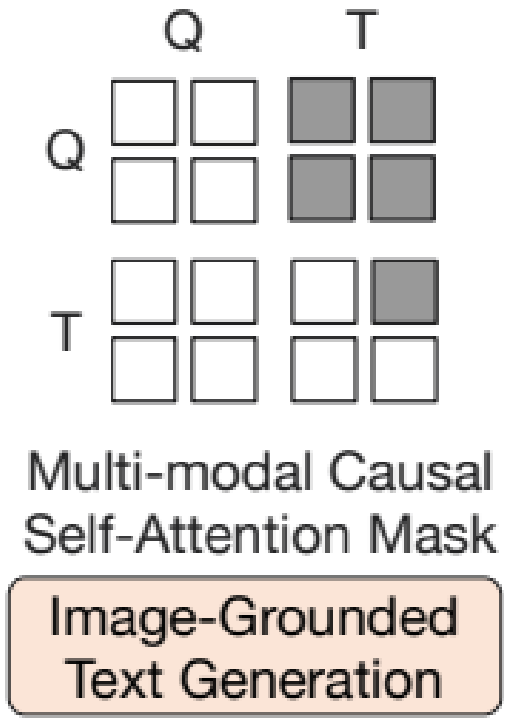


Figure 2: multimodal causal mask

Queries read each other and the image
Text reads all queries and its existing text prefix

$$\mathcal{L}_{\mathrm{ITG}} = -\sum_{t} \log p_\theta(y_t \mid y_{<t}, I)$$

“A cat wearing …” should predict “sunglasses”

Li et al., BLIP-2 (2023) · arxiv.org/abs/2301.12597

BLIP-2: Three Objectives and Attention Masks

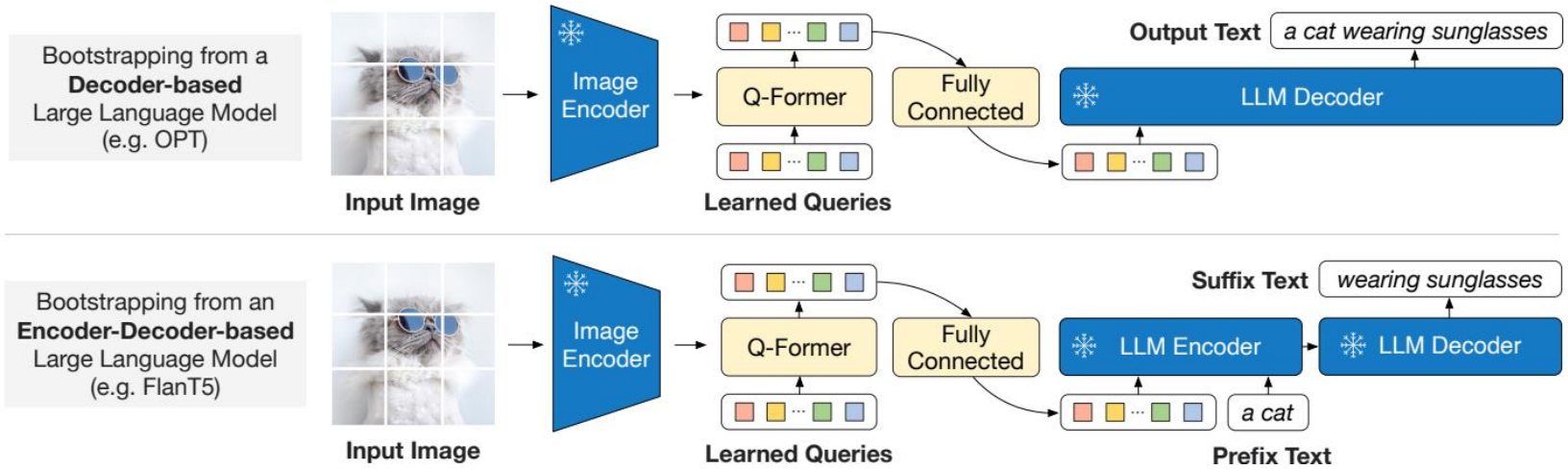
Objective	Query–text attention	What is rewarded
ITC: contrastive	Separated streams	Match paired image/text representations
ITM: matching	Bidirectional interaction	Distinguish matched from hard-negative pairs
ITG: generation	Text sees queries + earlier text	Predict caption words from visual queries

$$s(I,T) = \max_k \text{sim}(z_k, t), \qquad \mathcal{L}_{I\to T} = -\log \frac{\exp(s(I,T)/\tau)}{\sum_j \exp\left(s\left(I,T_j\right)/\tau\right)}$$

Shown: one direction of ITC; BLIP-2 also optimizes the reverse direction

Li et al., 2023 · https://arxiv.org/abs/2301.12597

BLIP-2: Connecting Queries to the Generator

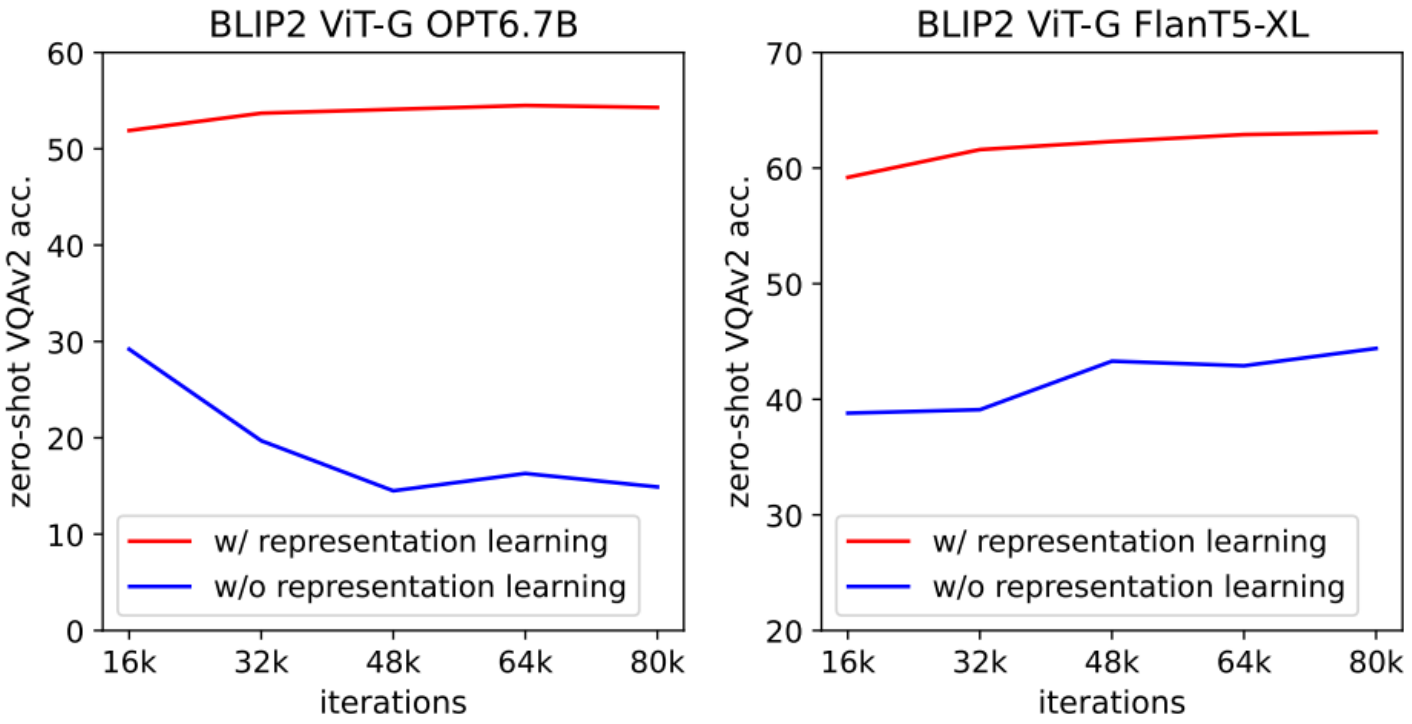


Original Figure 3: decoder-only and encoder–decoder variants

$$Z = W\, Q_\phi\big(E(I)\big), \qquad \mathcal{L}_{\mathrm{LM}} = -\sum_t \log p_\theta(y_t \mid y_{<t}, Z)$$

Li et al., 2023 · https://arxiv.org/abs/2301.12597

Does the First Pretraining Stage Matter?



Same generative-learning stage
Different initialization of Q-Former

Both tested LLMs benefit from representation pretraining

Evidence for this training recipe (budget, dataset, etc.); not a universal requirement for every connector

Li et al., BLIP-2 (2023), Figure 5 and §4.1

Principles of Multimodal Alignment

- Matching, pair discrimination, and text generation training
- Use attention masks to define different prediction tasks
- The connector learning problem can be decomposed into: feature selection and interface compatibility

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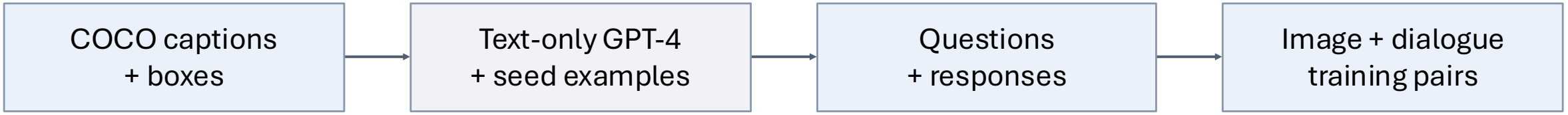
From Caption Completion to Task Specification

Instruction	Desired response
Describe the scene	A red mug is beside a blue bowl
What is left of the bowl?	The red mug
Name only the red object in one word	Mug

How do we obtain training examples for these different requests?

Teaching example · connection to Lecture 4: task specification

LLaVA: Constructing Instruction Data



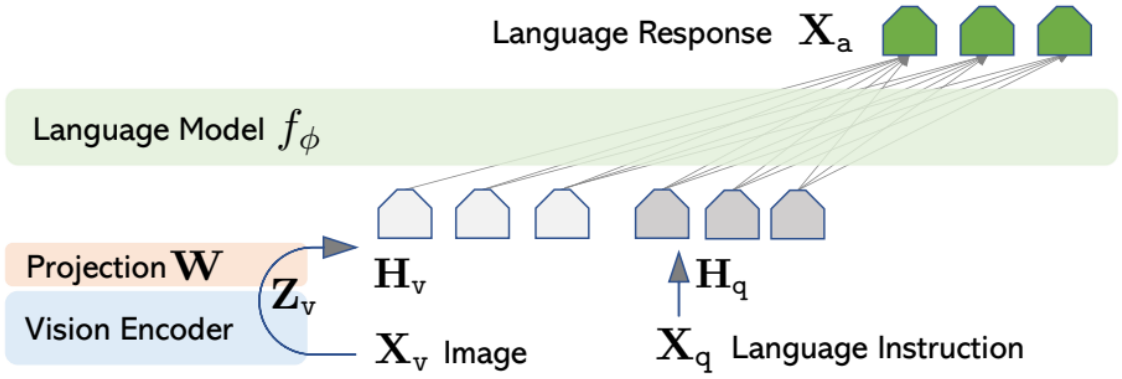
58K conversations **23K descriptions** **77K reasoning samples**

$$\mathcal{L}_{\text{inst}} = -\sum_{t \in S} \log p_{\theta,\phi}(y_t \mid I, q, y_{<t})$$

Risk: annotation omissions and teacher inferences become part of the supervision

Liu et al., 2023 · https://arxiv.org/abs/2304.08485

LLaVA: Learning to Follow Instructions



Original Figure 1: CLIP, linear projection, Vicuna

Key idea

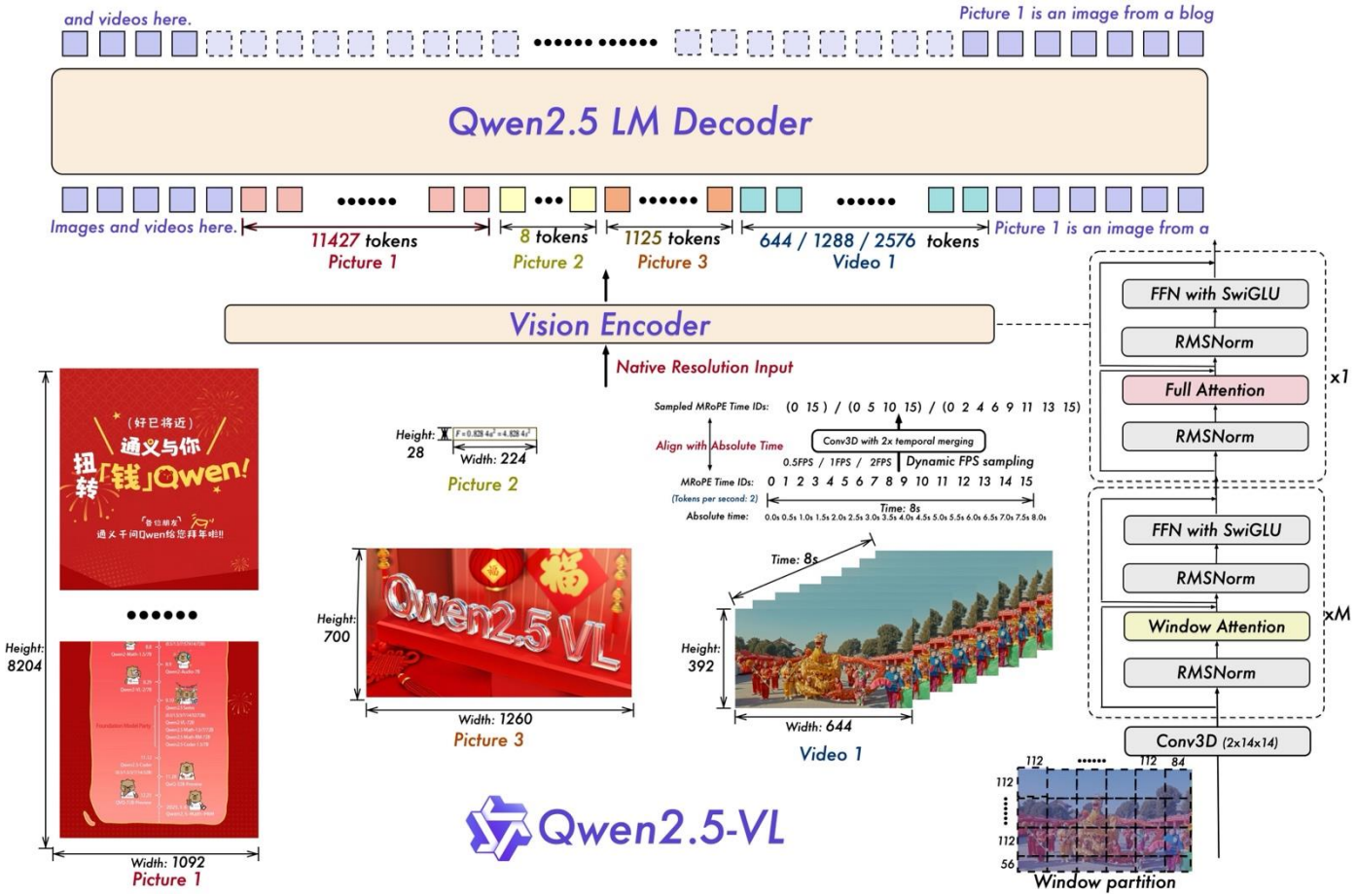
Teach image-conditioned instruction following with response supervision

Stage	Trainable	Supervision
1. Feature alignment	Projector	595K image-caption pairs
2. Instruction tuning	Projector + LLM	158K generated instruction samples

Original LLaVA: linear projection Later LLaVA-1.5: two-layer MLP

Liu et al., 2023 · https://arxiv.org/abs/2304.08485

Qwen2.5-VL: Scaling the Visual Interface



Original Figure 1: variable-length visual inputs

The same broad interface as LLaVA

Vision encoder + MLP merger
Visual tokens join the text sequence
Qwen2.5 decodes the response

Key design changes

Faster Vision Backbone
MRoPE: multimodal positional embedding

Qwen Team, 2025 · arxiv.org/abs/2502.13923

Qwen2.5-VL: Resolution and Token Budget

Local spatial merging

Resize within a pixel budget to multiples of 28, then encode 14 × 14 patches
Concatenate each 2 × 2 group of features and apply a two-layer MLP

$$N_{\text{visual}} = \frac{H}{28}\frac{W}{28}, \qquad 224\times224:\ 64 \quad \rightarrow \quad 448\times448:\ 256$$

Efficient vision attention

28 of 32 ViT layers use windows of at most 8 × 8 patches
Four global layers exchange information across windows

Qwen Team, 2025 · arxiv.org/abs/2502.13923

Qwen2.5-VL: Position in Space and Time

ViT uses 2D RoPE; the language model backbone uses MRoPE

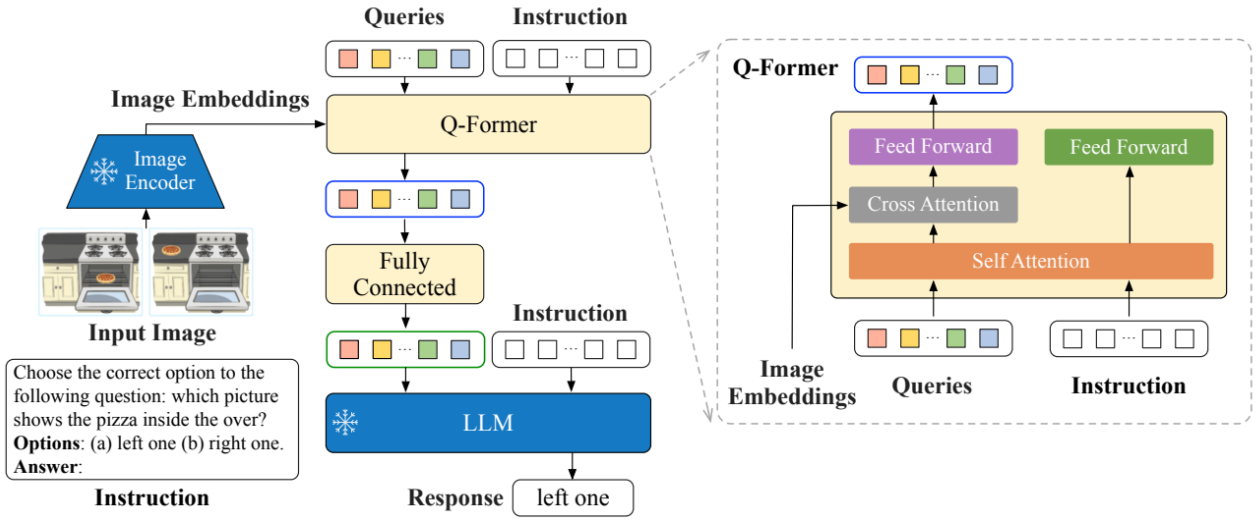
Split query/key channels for time, height and width, then rotate each part

$$\tilde{q} = [R(p_t)q_t \; ; \; R(p_h)q_h \; ; \; R(p_w)q_w], \qquad \text{likewise for } k$$

Text	All three position IDs advance together: ordinary 1D RoPE: (k, k, k)
Images	Time stays fixed, while row and column IDs follow the merged grid (k, i, j)
Videos	Temporal IDs = actual elapsed time, rather than the sampled-frame index Changing FPS changes observations but preserving the timestamp (t, i, j)

Qwen Team, 2025 · arxiv.org/abs/2502.13923

InstructBLIP: Instruction-Aware Extraction



Original Figure 3: instruction-aware Q-Former

$$Z = f(I, q)$$

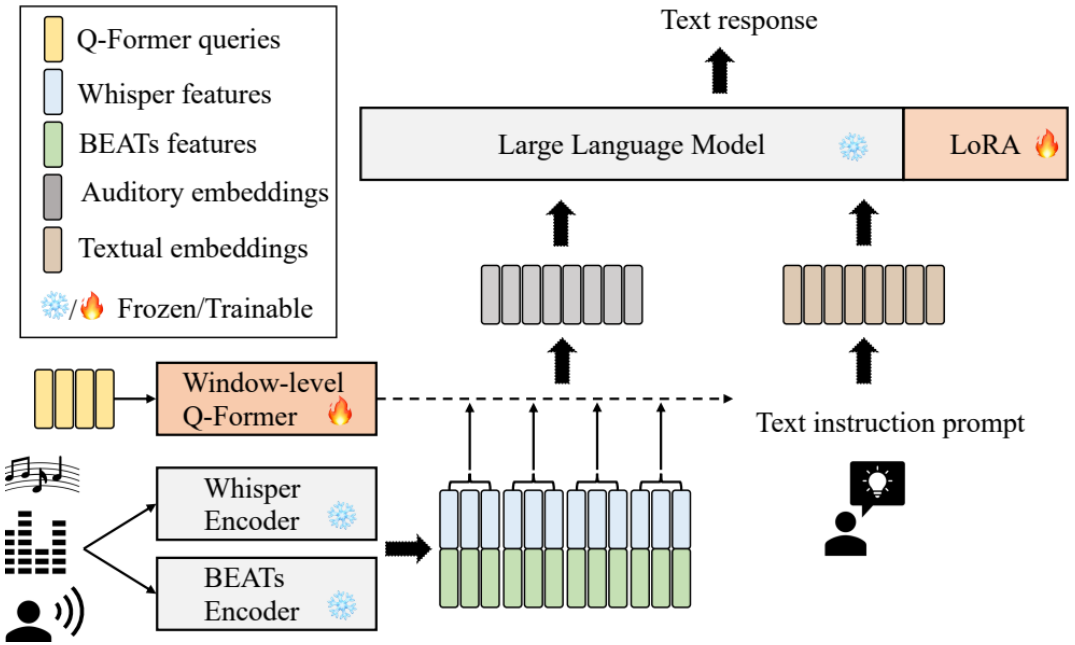
“What color is the mug?”
“Where is the knife?”

Same image, different information needs
13 training datasets, 13 held-out datasets

Key idea: let the instruction influence what the connector extracts

Dai et al., 2023 · https://arxiv.org/abs/2305.06500

SALMONN: Connecting Speech and Sound



Original Figure 1: dual encoders and window-level Q-Former

Key idea

Combine complementary auditory features

Train the window Q-Former + LoRA
Keep both encoders and base LLM frozen

$$K_{\text{audio}} = N_q \left\lceil \frac{T}{L_w} \right\rceil$$

T: frames Lw: window length Nq: queries/window

Tang et al., 2023 / ICLR 2024 · https://arxiv.org/abs/2310.13289

Principles of Multimodal Instruction Following

- Task specification needs training examples
- Instructions can guide what the connector extracts

Connecting Modalities to LLMs

- The Connector problem: extract useful information (Projection Layers, Q-Former, conditional Q-Former) and align the “interface”
- Issues with this view: the encoder should ideally be aware of what information to extract!
- Training objectives: contrastive (fast), auto-regressive generation (slow)
- Synthetic data generation!

Questions for Discussion

- In your domain, what naturally occurring data could connect a sensory encoder to an LLM? Can we generate more data by composing existing models?
- What information may a connector discard, and how would you detect the loss?
- How would you separate task-following errors (e.g., question-answer accuracy) from failures to use sensory evidence (e.g., object detection precision)?